Class 10 NCERT Maths

Chapter 1: Real Numbers - Detailed Guide

CLASS 10 NCERT MATHS
CHAPTER 1: REAL NUMBERS

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

PART 1: FOUNDATION (Class 7-8-9 ka revision)

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1. NUMBERS KE TYPES - EK FAMILY TREE

Sochh sabhi numbers ek bada family hai.

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              Real Numbers (R)
              /            \
       Rational (Q)       Irrational (P)
        /      \              |
  Integers (Z)  Fractions   sqrt(2), pi, e
     /   \      (1/2, 3/4)
 Whole(W)  Negative
  /   \   (-1, -2, -3)
Zero  Natural (N)
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(1, 2, 3, 4...)

DEFINITIONS EASY BHASHA MEIN:

- Natural Numbers (N): Counting numbers - 1, 2, 3, 4, 5...
- Whole Numbers (W):   Natural + 0   - 0, 1, 2, 3...
- Integers (Z):   Whole + negative - ..., -2, -1, 0, 1, 2...
- Rational Numbers (Q): p/q form (q != 0)- 1/2, -3/4, 5, 0.25
- Irrational Numbers: NOT p/q form   - sqrt(2), sqrt(3), pi
- Real Numbers (R):   Rational + Irrational (sab kuch)

2. RATIONAL VS IRRATIONAL - KAISE PEHCHANEN

RATIONAL number ki decimal expansion 2 type ki hoti hai:
- Terminating (khatam ho jaye): 0.5, 0.25, 0.125
- Non-terminating but Repeating: 0.333..., 0.142857142857...

IRRATIONAL number ki decimal expansion:
- Non-terminating AND Non-repeating
- Example: sqrt(2) = 1.41421356237...

3. PRIME NUMBERS VS COMPOSITE NUMBERS

PRIME NUMBER:

Wo number jo SIRF 1 aur apne aap se divide hota hai.
Examples: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29...

IMPORTANT:

- 1 prime NAHI hai!
- 2 ekloti EVEN prime hai (baaki sab prime odd hain)

COMPOSITE NUMBER:

Jo prime nahi hai (1 ke alawa). Iske 2 se zyada factors hote hain.
Examples: 4, 6, 8, 9, 10, 12, 14, 15...

EXAMPLES:

7 ke factors: 1, 7 -> 2 factors -> Prime
12 ke factors: 1, 2, 3, 4, 6, 12 -> 6 factors -> Composite

4. HCF AUR LCM - BACHPAN SE YAAD RAKH

HCF (Highest Common Factor) = GCD
Sabse bada number jo dono ko divide kare.

LCM (Least Common Multiple)
Sabse chhota number jo dono se divide ho.

EXAMPLE: 12 aur 18 ka HCF aur LCM
12 ke factors: 1, 2, 3, 4, 6, 12
18 ke factors: 1, 2, 3, 6, 9, 18
Common factors: 1, 2, 3, 6
HCF = 6 (sabse bada common)

Multiples of 12: 12, 24, 36, 48...
Multiples of 18: 18, 36, 54...
LCM = 36 (sabse chhota common)

MAGIC FORMULA:

HCF x LCM = Product of two numbers

Check: 6 x 36 = 216
12 x 18 = 216
Match!

PART 2: ASLI CHAPTER 1 CONTENT (NCERT)

TOPIC 1: THE FUNDAMENTAL THEOREM OF ARITHMETIC

STATEMENT:

"Every composite number can be expressed (factorised) as
a product of primes, and this factorisation is UNIQUE,
apart from the order in which the prime factors occur."

AASAAN BHASHA MEIN:

Koi bhi composite number ko prime numbers ke multiplication
se likh sakte hain, aur ye tareeka UNIQUE hota hai.

EXAMPLE 1: 156 ka prime factorisation kar

156 / 2 = 78
78 / 2 = 39
39 / 3 = 13
13 / 13 = 1

So, 156 = 2 x 2 x 3 x 13 = 2^2 x 3 x 13

EXAMPLE 2: 3825 ka prime factorisation

3825 / 3 = 1275
1275 / 3 = 425
425 / 5 = 85
85 / 5 = 17
17 / 17 = 1

So, 3825 = 3^2 x 5^2 x 17

EXAMPLE 3: 7 x 11 x 13 + 13 - kya ye composite hai?

= 13 x (7 x 11 + 1)
= 13 x (77 + 1)
= 13 x 78

Haan, ye composite hai kyunki iske factors
1, 13, 78 ke alawa aur bhi hain.

TOPIC 2: HCF AND LCM USING PRIME FACTORISATION

METHOD:

HCF = Common prime factors ka product (LOWEST power)
LCM = Sab prime factors ka product (HIGHEST power)

EXAMPLE 4: 96 aur 404 ka HCF aur LCM nikaal

Step 1: Prime factorisation
96 = 2^5 x 3
404 = 2^2 x 101

Step 2: HCF (common primes, lowest power)
Common prime: 2 (lowest power = 2^2)
HCF = 2^2 = 4

Step 3: LCM (all primes, highest power)
LCM = 2^5 x 3 x 101 = 32 x 3 x 101 = 9696

Verify: HCF x LCM = 4 x 9696 = 38784
96 x 404 = 38784
Match!

EXAMPLE 5: 6, 72, 120 - teen numbers ka HCF aur LCM

Prime factorisation:
6 = 2 x 3
72 = 2^3 x 3^2
120 = 2^3 x 3 x 5

HCF (common, lowest power):
2^1 x 3^1 = 6

LCM (all, highest power):
2^3 x 3^2 x 5 = 8 x 9 x 5 = 360

TOPIC 3: REVISITING IRRATIONAL NUMBERS

THEOREM:

Let p be a prime number. If p divides a-square (a^2),
then p divides a (where a is a positive integer).

FAMOUS PROOF: sqrt(2) IS IRRATIONAL
(Ye exam mein aata hai!)

Proof by CONTRADICTION:

Step 1: Maan le sqrt(2) rational hai.
Toh sqrt(2) = p/q
where p, q are integers, q != 0,
and HCF(p, q) = 1 (simplest form).

Step 2: Dono sides square karo
2 = p^2 / q^2
2 q^2 = p^2 ...(i)

Step 3: Iska matlab 2 divides p^2.
Toh theorem se, 2 divides p.

Step 4: Toh p = 2c (kisi integer c ke liye).

Step 5: Equation (i) mein daal
2 q^2 = (2c)^2 = 4 c^2
q^2 = 2 c^2

Step 6: Iska matlab 2 divides q^2,
toh 2 divides q.

Step 7: Ab problem!
Humne kaha tha HCF(p, q) = 1,
par dono ko 2 divide kar raha hai.
Ye CONTRADICTION hai.

CONCLUSION:

Hamari assumption galat thi.
Toh sqrt(2) IRRATIONAL hai. (Proved)

EXAMPLE 6: Prove that sqrt(3) is irrational

1. Maan le sqrt(3) = p/q (HCF = 1)
2. 3 q^2 = p^2 -> 3 divides p^2 -> 3 divides p
3. p = 3c -> 3 q^2 = 9 c^2 -> q^2 = 3 c^2
   -> 3 divides q
4. Contradiction (HCF = 1 ko hum break kar diye)
5. So sqrt(3) is irrational.

EXAMPLE 7: Prove that 5 - sqrt(3) is irrational

1. Maan le 5 - sqrt(3) rational hai, kaho r.
2. Toh sqrt(3) = 5 - r.
3. 5 - r rational hai (rational - rational = rational).
4. Par sqrt(3) irrational hai.
5. Contradiction! So 5 - sqrt(3) is irrational.

EXAMPLE 8: Prove that 3 x sqrt(2) is irrational

1. Maan le 3 x sqrt(2) rational hai, kaho r = p/q.
2. Toh sqrt(2) = p / (3q), jo rational hai.
3. Par sqrt(2) irrational hai.
4. Contradiction! So 3 x sqrt(2) is irrational.

USEFUL RULES TO REMEMBER:

- Rational + Irrational = Irrational
- Non-zero Rational x Irrational = Irrational
- Irrational + Irrational = could be either
  (e.g., sqrt(2) + (-sqrt(2)) = 0 which is rational)

PART 3: PRACTICE PROBLEMS (Tu solve kar)

Q1. Express 140 as a product of its prime factors.

Q2. Find the HCF and LCM of 26 and 91 by prime factorisation.

Q3. Prove that sqrt(5) is irrational.

Q4. Show that 3 + 2 x sqrt(5) is irrational.

Q5. Find LCM and HCF of 17, 23, 29.

SUMMARY (YAAD RAKH)

1. Fundamental Theorem of Arithmetic:
   Har composite number = UNIQUE product of primes.

2. HCF x LCM = Product of two numbers
   (sirf 2 numbers ke liye, 3 ke liye nahi).

3. HCF = common primes x LOWEST powers

4. LCM = all primes x HIGHEST powers

5. sqrt(p) is IRRATIONAL when p is prime.

6. Rational + Irrational = Irrational

7. Non-zero Rational x Irrational = Irrational

TIPS FOR EXAM

1. Prime factorisation hamesha tree method se kar - mistake nahi hogi.

2. HCF aur LCM ke questions mein dono nikalna hota hai. Verify karne ke liye HCF x LCM = product formula use kar.

3. Irrationality proofs mein "Proof by Contradiction" ka exact format yaad rakh:
   - Assume opposite
   - Derive equation
   - Apply theorem
   - Find contradiction
   - Conclude

4. Common irrationality questions:

sqrt(2), sqrt(3), sqrt(5), sqrt(7) - prove karne aate hain
a + b x sqrt(p) form ke numbers - prove karne aate hain

5. Numericals practice kar - NCERT exercise 1.1 aur example questions saari solve kar.
